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Deep Eutectic Liquids as Tailorable Extraction Solvents: A Review of Opportunities and Challenges

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ABSTRACT

Deep Eutectic Liquids (DELs) fall among the rapidly evolving discoveries of the 21st century, and these liquids are considered as alternative solvents to toxic and volatile organic liquids. Nevertheless, the emerging trend regarding the use of DELs in every field of physical and biological sciences, a lot of ambiguities and misconceptions exist about their formation, mechanism, and efficiencies observed or projected. A review of available technical data makes it obvious that these liquids have the potential to revolutionize the underdeveloped areas of analytical chemistry particularly the extraction/enrichment of analytes. To ensure the green and sustainable use of DELs, the researchers need to have a thorough understanding of DELs, their classification, chemistry, the nature and strength of molecular entanglements, and their tailorable features. Many researchers have declared these liquids recyclable but more attentive trials are needed to develop an authentic and straightforward DELs recycling methodology. The present review covers sound background knowledge and expert opinions about the technical definition of DELs, their classification, formation, recyclability, and tailorable features for their application as extraction solvent/sorbent in analytical chemistry.

KEYWORDS

Analytical chemistry; deep-eutectic liquids; extraction solvents; tailorable and green

Deep eutectic liquids

Firstly noticed in 2001,[1] and then coined as "deep eutectic solvents" in 2003 by Abbott et al.,[2] these liquids have the potential to revolutionize the fields of environmental, polymer, biological, and material sciences. In Greek, the term "ευτηκτος" is pronounced as "eutectic" which means easy/well melting,[3,4] and henceforth it denotes a mixture that usually becomes liquid (low isobaric melting temperature) at room temperature. Actually, the term "Deep" has its own technical significance in the formation and utilization of these liquids. This term infers a mixture that liquifies prior to its ideal eutectic point, hence if the mixture of various solids becomes liquid before its ideal eutectic point should be labeled as a deep eutectic liquid. According to the strict definition of the deep eutectic point, the mixture which becomes liquid at room temperature cannot be labeled as deep eutectic liquid unless it melts below the ideal melting point. Martins et al.[5] already suggested that using the term "liquid at room temperature" is a vague notation and it should be wiped out. Another misconception that needs to be highlighted here is the use of the term "solvents" for these liquids. The word "solvent" can only be assigned to a deep eutectic liquid if it finds its applications only as a solvent but the literature shows that many deep-eutectic liquids are equally useful as drug careers, electrolytes, and biocatalysts. Moreover, in the majority of analytical reports, the deep-eutectic liquids have been integrated with conventional liquids like water, ethanol, and methanol, before their use as an extraction or synthesis solvent. We have, therefore, decided to use the more broader and universal term (Liquids) i.e., "deep eutectic liquids" instead of "deep eutectic solvents" in this article, and the same should be adopted worldwide as many of the deep eutectic liquids (DELs) do not find their applications as solvents. Recently, Şahin[6] also suggested this correction regarding the labeling of these tailorable liquids. Another important reason behind suggesting this name is their resemblance with synthetic liquids, for which we apply the term ionic liquids (ILs) rather than ionic solvents (ISs). It is also essential to remember that DELs are not pure compounds but consist of non-reacting, reversible, and isothermal mixtures of different chemical species which may exist as a liquid at low temperatures. A similar misconception that needs to be highlighted and corrected here exists regarding the formation of deep-eutectic liquids. In a few research reports published regarding the formation of DELs the term "synthesis"[7–10] has been used which in fact denotes the chemical reactions but the formation of DELs does not involve any chemical reaction rather it happens due to the reversible, isothermal, and physical interaction between various species. The mixture prepared under a specific stoichiometric ratio may have freezing points lower than either of the two components. Therefore the term

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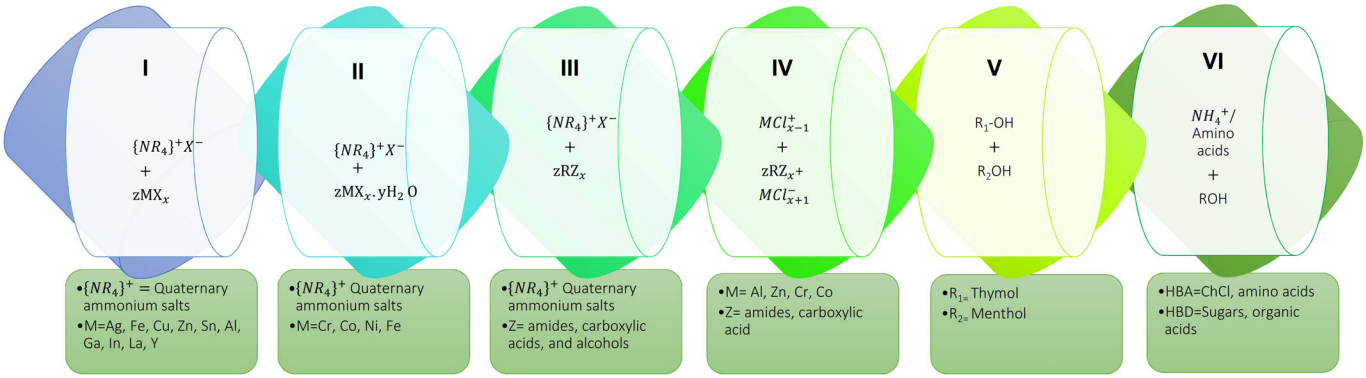


Figure 1. Types of deep eutectic liquids (DELs) with increasing green features (L-R).

"preparation", "formation" or "development" should be adopted instead of "synthesis" in the works undertaken to produce these liquids.

The presence of strong hydrogen bonds and/or other intermolecular interactions render most of these liquids nonvolatile, non-flammable, and sometimes liquid at room temperature. The principles of green chemistry[11] categorize these liquids as green liquids mainly for their nonvolatile, non-flammable, and nontoxic character. The deep eutectic liquids may rank as one of the most important discoveries of the 21st century. These liquids have now become potential candidates to replace toxic conventional organic solvents and expensive ionic liquids (IL) for analytical, synthesis, and electrolytic applications. Besides, their low cost, ease of raw material availability, biodegradability, and biocompatibility make them more viable than ionic liquids. Another citeworthy opportunity regarding these liquids rises due to change in their physicochemical properties with the variation in the mixing ratios of Hydrogen Bond Acceptor (HBA) and Hydrogen Bond Donor (HBD) species.[12,13] It is fair to assume that millions of different possible combinations can be designed by changing the types of HBD/HBA species and their respective mixing ratios. Recently, Cunha and Fernandes[14] compared the effectiveness of DELs in sample preparation for analytical chemistry, however, this review does not provide an insight into the key physicochemical features that affect their solvation power. The present review aims to highlight the designable physicochemical features of deep eutectic liquids (DELs). The monograph also contains comprehensive information and expert opinions regarding the interactions between HBD and HBA, opportunities, and challenges which should be addressed and publicized before their widespread utilization as extraction solvents.

Before we proceed toward tailorable features and utilization of DELs as extraction solvents, it is critical to consider the technical definition of DELs, their classification, characterization, and the fundamental mechanism involved in their formation. Once became familiar with the molecular interactions in deep eutectic liquids and the components responsible for such interactions, we can customize the physicochemical parameters of these liquids and harmonize them for more intelligent use in various fields.

Classification of deep eutectic liquids

It had been generally believed that transition metal salts would not ionize in organic solvent until 2001 Abbott et al.[1] found that $ZnCl_2$ can form a eutectic mixture with urea, ethylene glycol, acetamide, and 1,6 hexanediol. The deep eutectic liquids (DELs) fall into five major classes with respect to the nature of HBD and HBA involved (Figure 1) in their formation. The most frequently reported and the first generation of DELs (labeled as type-I) was prepared by the combination of quaternary ammonium salts (HBA) and metal chlorides (HBD). The members of type-I DELs are difficult to prepare and rarely used as extraction or synthesis solvent for their toxic, and non-biodegradable nature.[11] The next generation (Type II) of DELs compose of similar HBD and HBA, nonetheless, metal chlorides are hydrated to increase the scope and range of produced combinations. Another good perspective of type II DELs rises due to the ease of preparation as metal chlorides are directly used with their inherent air/moisture. The III type of DELs also contains similar quarternary ammonium compounds like choline chloride (ChCl) as HBA, but here HBD may be alcohol, amine, or carboxylic acid.[15] These DELs are easy to prepare, economical, and non-reactive toward biomolecules. Besides, the abundance of organic hydrogen bond donors offers a wide range of combinations, adaptability, and applications for this class of liquids.

The most interesting class of DELs (type IV) has been prepared by mixing quarternary ammonium compounds (HBAs) with hydrogen bond donors (HBDs) like urea, alcohols, and carboxylic acids. Some researchers have also prepared type IV DELs by mixing HBDs and metal chloride hydrates.[16] These DELs are generally hydrophilic (due to the ionic components), which limits their use in aqueous samples/medium only. The curious case of nonionic or hydrophobic DELs has served as a possible solution in which strong abnormal interactions are developed between the components of type V of DELs.[17] These hydrophobic DELs can be further classified into two groups i.e., (i) ionic and (ii) nonionic. In most of the cases, only long-chain fatty acids, alcohols, and monoterpenes having hydrophobic components work well to prepare these liquids. Recently, Choi et al.[18] prepared natural deep eutectic liquids (NaDELs)

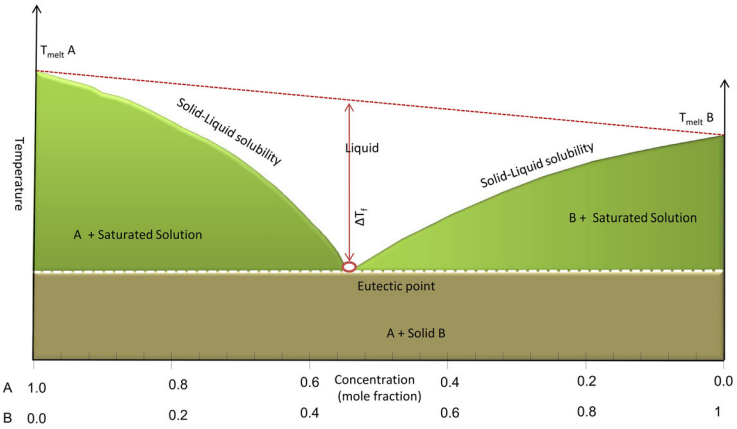


Figure 2. A typical liquid-solid phase behavior of a binary mixture that intercalates at a particular stoichiometric ratio. The DELs involve a further unexpected deep depression of freezing points at particular mole ratios.

from natural HBDs and HBAs and claimed that many of such combinations already exist in living cells and provide a natural medium for various physiological phenomena.[18] A growing trend has appeared regarding the preparation and utilization of NaDELs comprising amino acids as HBAs while sugars and various organic acids as HBDs for their biocompatibility and biodegradability. The NaDELs also offer a wider polarity range, low volatility, and high solvation power.[19] Besides, the NaDELs prepared by mixing organic acid and glucose at different ratios (Figure 1) are nontoxic and biocompatible. Another extensively studied version of type IV DELs include magnetic deep-eutectic liquids (MDELs) made up of magnetic nanomaterial, carbon nanotubes, metal organic framework (MOF) or paramagnetic anions like $[FeCl_4]^{-1}$.[20] The MDELs have been frequently applied as synthesis solvent as their physicochemical features vary in response to applied magnetic fields, scarce studies focused on their use as a solvent or sorbent.

Formation of deep eutectic liquids

As stated earlier, deep eutectic liquids (DELs) are customized mixtures of two or more species that exist as a liquid, so their preparation involves much simpler and easy steps as compared to their counterparts (ionic liquids). In general, the HBD and HBA species in a particular stoichiometric ratio are mixed, agitated, and/or heated under a vacuum, and then cooled until the interaction between HBD and HBA species becomes dominant. The exact protocol may vary with the nature (melting and boiling points, and stabilities) of HBD and HBA species, personal preference, the extent of available technologies, and end-user applications. In any case, prolonged and abrupt heating may degrade the constituents to limit the end-user application's effectiveness. Likewise, the presence of moisture in any cause can affect the formation and physicochemical attributes of the subsequent DEL system.

Overall, DELs have been prepared via three standard procedures, *i.e.*, (i) continuous heating of the components having fixed molar ratios in a closed flask at 50–90 °C with constant agitation for 30–90 min till the mixture becomes a homogenous transparent liquid as shown in Figure 2.[19] The presence of moisture or impurities in any of the HBD, and/or HBA species drastically change the eutectic point and related physicochemical properties. Therefore, (ii) vacuum drying the components first in a rotary evaporator at 50 °C followed by mixing them to get transparent liquids has been found more suitable for the preparation of DELs which often involve liquid HBD and HBA species. Drying the solid HBD or HBA species in a desiccator with silica or silica gel[21] also works fine but has been declared more time-consuming. Freeze-drying is another way (iii) to prepare DELs and it involves mixing of HBD and HBA species in a stoichiometric ratio, followed by dilution with distilled water to get transparent solutions that are subsequently freeze-dried for 2–24 h.[22] The concerns are there if the DELs produced via this technique comply with the definition of deep-eutectic point. Neverthless, the liquids prepared via the freeze-drying approach often contain micelle, vehicles, or nonreactor capsules, rendering them more suitable for biological applications.

In addition to these, mechanical approaches involving grinding the HBD and HBA species in mortar and pestle and extrusion using Twin Screw Extruder (TSE) have also been applied for the formation of DELs.[10] The TSE technique has been announced to be efficient and rapid to produce high-purity DELs. During most of the mechanical methods, the HBD and HBA species are dried under vacuum until there is no further weight loss and then subjected to grinding or extrusion to get a clear low melting solution. This strategy also seems to be more suitable for thermally labile HBD and HBA species.[7,10] Gomez et al.[23] have applied microwave heating to form low melting liquids and noted that microwave heating with irradiation power of 200 W when applied for only 20 seconds could produce low melting liquids and reduce energy and time consumption 650 and 180 times, respectively. Likewise, Santana et al.[24] have used ultrasound waves for the green formation of these liquids.

Structural characterization of DELs

Even though the synthetic or structural characterization of DELs does not contribute to the end-user application unlike

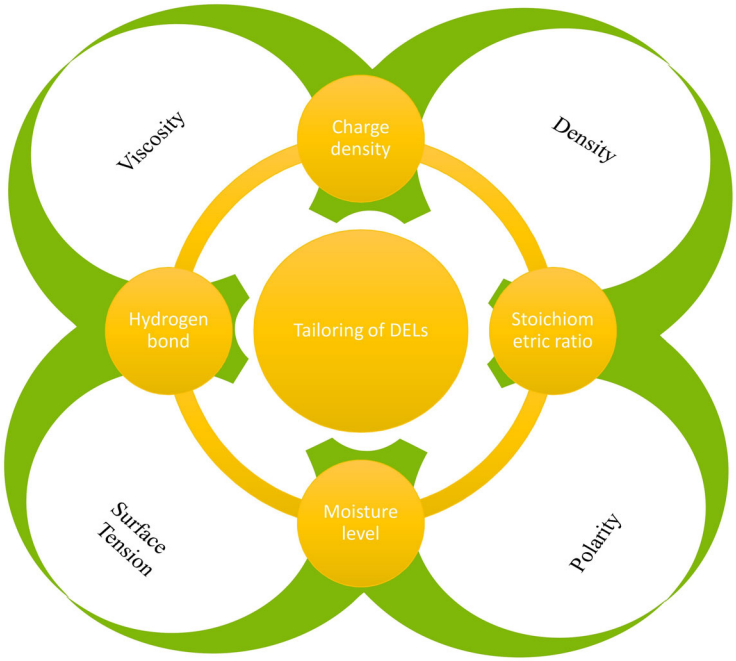


Figure 3. The most critical features of deep eutectic liquids (DELs) and tailoring approaches.

the physiochemical and thermodynamic characterization. The structural characterization of DELs has been considered one of the most authentic ways to confirm the formation of a particular DEL system and the majority of researchers have followed variations in physiochemical parameters to confirm the formation of DELs. Some researchers[25,26] have applied differential scanning colorimetry (DSC) while many others used simple cryoscope.[27] In DSC, 10–20 mg of liquids have been taken in aluminum pans and cooled to various temperatures followed by heating at a particular ramp rate to monitor various thermal transitions. The presence of endotherms at different temperature and their comparison with the DSC thermogram of parent HBD and HBA species can help us to establish the eutectic point of DEL and the presence of heterogeneity (if any). Besides, theoretical models like molecular dynamics (MD) and radial distribution function (RDF) have successfully predicted the number of hydrogen bonds and linkage between various atoms.[26]

Likewise, the presence of hydrogen bonds formed due to HBD and HBA are of special character and can be easily detected by nuclear magnetic resonance (NMR). For this purpose, the NMR spectra of HBD, HBD, and DEL are often compared. For example, the DEL system comprising ChCl:*n*-butyric acid when dissolved in DMSO-d_6 exhibited an additional peak at 6.93 ppm that indicates the appearance of intermolecular hydrogen bonds.[28] The infrared spectroscopy under Fourier-Transformation mode (FT-IR) stands among the most convenient and frequently applied techniques for DEL characterization. For every DEL system the appearance, disappearance, and broadening of peaks in the range between 1500 and 4000 cm^{-1} establish the formation of the resultant DEL. For example, the OH signal of propylene glycol becomes broader due to the formation of hydrogen bonds.[29] Unlike, the disappearance of signal of the methyl group (3025 cm^{-1}) of ChCl indicates the interaction of this methyl group and another. Recently, Cui and Kuroda[26] applied two dimensional infrared (2DIR) to probe into the variation in molecular environment and entanglement between the location and orientation of various groups. The 2DIR can directly monitor the difference in molecular arrangements of HBDs and HBAs.

Phase behavior

The key features of any DEL system depend on the eutectic composition and nature of hydrogen bond donors and acceptors. There exist two logics regarding deep depression at the freezing point (beyond a eutectic point). The first considers the eutectic point as a colligative effect and the second says crystallization may seize due to intercalation of HBDs into HBAs, and the resulting combination has the lowest possible melting point. The phase diagram in Figure 3 depicts the depression in the melting temperature of B due to the Intercalation of A and vice versa. In the case of deep eutectic liquids (DELs), an unexpected deep depression of the freezing point takes place for the simultaneous development of different interactions between HBA and HBD. According to Zhekenov et al.,[15] in the formation of choline chloride (ChCl) containing DELs, where ChCl act as HBA and urea (U), ethylene glycol (EG), and Glycerol (GY) act as HBDs, a decrease in the number of hydrogen bonds and intermolecular interaction between pure components results in the formation of low melting liquids. In particular, the interaction energy of ChCl before and after mixing decreases significantly. Moreover, the number of interactions and their resultant strength in the final low melting liquids vary with HBD. For example, EG and GY may attract anion with -5.0×10^4 and -3.5×10^4 kJ/mol energies, respectively, whereas, urea has a weak interaction with anion.

The intercalation often occurs when HBD and HBA of compatible size, charge, and nature are mixed in a particular ratio. Therefore, eutectic points can furnish important information about the composition, temperature range, and physiochemical characteristics of any eutectic system for a particular application. For example, ChCl and glycerol melt at 281, 237, and 241 K when mixed in the stoichiometric ratios of 1:1, 1:2, and 1:3, respectively.[30] Similarly, the physiochemical properties like melting points, surface tension, viscosity, refractive index, and pH change with the change in molar composition.[31] Researchers have also tried to identify the eutectic compositions by changing the compositions of HBA and HBD.[32]

A cite worthy break-through regarding the hydrophobicity of deep eutectic liquids has been documented by Longeras et al.[33] These researchers prepared the aqueous solution of a deep-eutectic liquid comprising lidocaine and oleic acid (1:1) and observed that a 50/50 mixture of water and deep-eutectic liquid form a single transparent liquid phase at 20 °C. The phase separation occurred at 25 °C (the lower critical solution temperature of the prepared deep eutectic liquid). It was further noted that an interface between water and the deep eutectic mixture of lidocaine and oleic acid appears once the temperature rises to 25 °C and disappears below 24 °C. The components of deep eutectic liquid can accumulate in the hydrophobic phase and move exclusively toward the aqueous phase. These types of deep eutectic liquids may work more efficiently to extract pollutants, dyes, and other bioactive substances.

A mathematical relationship between eutectic composition and enthalpy of transition can be produced by following Equation 1. Here X_i^L, ΔH_i, and Tm_i stand for mole fraction, enthalpy, and the melting point of the pure compound.[34]

$$\mathrm{Ln}X_i^L = -\frac{\Delta H_i}{RT}\left\{1 - \frac{T}{Tm_i}\right\} \quad (1)$$

For many combinations, these systems are also labeled as designer liquids that can be utilized more effectively by studying various other compositions apart from the eutectic one due to their changing dynamics and properties. Besides, the formation of deep-eutectic liquids often follows the first-order kinetics (Equation 1) for the activation parameters. For example, the formation of ChCl:urea, the most famous DEL, is an endothermic process with an enthalpy change of about 30 kJ mol^{-1}. Applying the Arrhenius equation establishes low activation energy (up to 11 kJ mol^{-1}) and a more prominent pre-exponential factor for the ChCl:urea. Likewise, positive entropy of formation and negative Gibbs function indicates the process is reasonably fast.

Hydrogen bonding

The classical deep eutectic liquids (DELs) are low melting mixtures of general formula $[NPS]^+X^-NY$, where NPS are organic cations of ammonium, phosphonium, or sulphonium species carrying transferable Lewis base, *i.e.*, Halide anion, and Y represents Hydrogen bond donors. However, certain studies describe the halogen-free liquids mainly comprising alkali acetate, triflate, and bistriflimide as HBA and polyalcohols as HBD.[35] The main forces of interaction holding the components of deep eutectic liquids together are hydrogen bonds; however, π- interactions and Van der Waal forces are also notified. Overall, a cascade of hydrogen bonds, i.e., weaker, moderate, and stronger may exist between HBD and HBA species depending upon their inherent polarity.

The key factor behind the lowering of melting point in any DEL apart from ionic and steric effects is hydrogen bonding. According to Arunan et al.,[36] there is a charge transfer between HBA and HBD molecules as a result of which, a partial covalent bond is formed between the two molecules which can be identified via NMR, or FTIR. Pimentel and McClellan[37] highlighted the impact of hydrogen bonding on various properties like melting point, density, vapor pressure, and viscosity of DEL systems. The authors also noticed that a change in H-bond interactions is related to the change in density as the increase in density would decrease the free volume. The number of hydroxyl groups at vicinal position often lead to intracts within the molecules of HBDs themselves. For example, Ozturk et al.[38] observed that the glycerol-containing DEL system exhibited higher density due to a higher number (three) of hydroxyl groups available for H-bonding. The DEL system containing ethylene glycol also exhibit higher density at the eutectic point than pure HBD due to the enhancement of H-bonding. In contrast, the glycerol-based DEL system exhibited minimum density at the eutectic point due to disruption of H-bonding. This creates serious interest in the role of H-bonding for a DEL system as both combinations had the same HBA. However, the difference in H-bonding compelled the systems to follow completely different mechanisms.[38] Stefanovic et al. reported that a similar H-bond pattern was present in DEL systems based on ChCl, ethylene glycol, and glycerol, while DEL containing ChCl and urea exhibited a unique pattern that differs in hydrogen-bonded networks.[39] Ashworth et al.[40] studied hydrogen bond interaction between components ChCl and urea containing DELs and reported that there happen different type of hydrogen bond interactions (substantial, regular, and weak) between the components, and the bonding is somewhat transient, which means that the molecules can switch their sites whenever they want to. This explains the anomalous behavior of DEL systems regarding glassy behavior and high viscosities.[40] These studies highlight that H-bonding plays a pivotal role in each DEL combination.

Tailorable features of deep eutectic liquids

A variety of electrostatic interactions no doubt complicate the assessment, but at the same time, it generates an equal number of polarity, viscosities, surface tension, and density arrangements. Figure 3 manifests the tailorable physicochemical properties of DELs and it would be fair to imagine the hydrogen bonding network of DEL resembles the

structure of spider's webs where the strength of each thread and their connectivity affects these physicochemical attributes.

Polarity

Polarity plays the most crucial role in deciding the depth of eutectic point (melting temperature, vapor pressure, and other physicochemical parameters associated with DELs), especially for those designed to replace the organic solvents being used in industry and academia. In the case of organic solvent, polarity expresses its ability to solvate various species and facilitate their dissolution or solvation. The polarity of DELs depends upon the cascade of bonding forces (H-bond, van Der Waal's interaction, or π-interaction), however, it is difficult to define and almost impossible to assess absolutely. A set of assumptions and methods has been followed to express the polarity in terms of the solvatochromic potential of the DELs, it should be mentioned here that these paradigms also depend on mass transfer rates. Pandey et al.[41] followed the UV visible and molecular fluorescence to probe into polarity-induced vibronic bands. For example, the absorption band of betaine dye 30 (2,6-diphenyl-4-(2,4,6-tri phenyl pyridinium-1-yl)phenolate) shifts from 810 nm (λ_{max}) in non-polar solvent to 453 nm (λ_{max}) in water (polar solvent) due to a charge transfer reaction between phenolate and pyridinium part. The zwitterionic nature of dye has been frequently exploited as a solvatochromic probe, and the presence of hydrogen bonding donating species stabilizes the ground state of dye. An empirical scale (ET_{30}= 28591.5/λ_{max}) expresses the molar transition energy (kcal mol^{-1}) at standard temperature and pressure. However, owing to the solubility restrictions, other probes have been developed. Almost all photo-inducible charge-transfer probes depend upon the hydrogen bonding ability of DELs and the strength of interaction between HBD and HBA.

Wojeicchowski et al.[42] studied the Kamlet-Taft (K-T) parameters associated with the solvatochromic potential of solvents. The K-T model parameters describe the polarizability (π^*), hydrogen bond accepting ability (β), and hydrogen bond denoting ability (α). After calculating the properties, including (a) misfit interaction energy (E_{mf}), (b) van der Waals interaction energy (E_{vdW}), hydrogen bond accept moment (M_{HBA}), and hydrogen bond denote moment (M_{HBD}) for 175 organic solvents, the approach was extended to various types of DELs following Equations 2–4.

$$\pi^* = X_1 M_{HBA} + X_2 M_{HBD} + X_3 E_{mf} + X_4 \quad (2)$$

$$\beta = X_1 M_{HBA} + X_2 M_{HBD} + X_3 E_{vdW} + X_4 \quad (3)$$

$$\alpha = X_1 M_{HBA} + X_2 M_{HBD} + X_3 E_{vdW} + X_4 \quad (4)$$

Finally, comparing predicted values of K-T model parameters for organic solvents and deep eutectic liquids indicates that the latter can have a broader range of polarities. Likewise, Florindo et al.[43] observed that the structure of HBA in a DEL system controls the polarity/polarizability, and K-T model parameters remain almost the same within a particular family of DEL; however, these parameters differ among the various families of DEL, whether they are made up of salts like ChCl, tertiary amines, acids or neutral compounds. Similarly, the substitution of HBD often changes the solvatochromic behavior of the resultant DEL.

Viscosity

The higher viscosities of DELs are a key hurdle in their use for various applications, particularly in extraction technology. The relative strength and number of hydrogen bonds shift the melting point deep below the eutectic point and the same can increase the viscosity of the resultant liquid. However, the geometry, shape, and size of molecules also contribute to their resistance in flow dynamics. It has been noted that pure DEL offers more resistance to change in shape at a given shear rate. The viscosity decreases with the increase in temperature and moisture level. The effect of temperature on the viscosities of DEL has been extensively studied and found to be testable by Arrhenius (Equation 5) and the (Vogel – Fulcher – Tammann) VFT (Equation 6) model.[44,45]

$$\eta Arrhenius = Ae^{E/RT} \quad (5)$$

Here, E represents activation energy, R is the molar gas constant, and A is a pre-factor.

$$\eta_{VFT} = A' e^{B/(T-To)} \quad (6)$$

where A′ represents the pre-exponential factor that quantifies viscosity, B accounts for activation energy, and T_o corresponds to the glass transition temperature. Keeping these equations in mind, a wide range of studies have been performed to quantify the effect of various factors like mole fractions of HBD/HBA and water content (%). Aroso et al.[46] analyzed various combinations of ChCl and sugars via rheology. As modeled by the Arrhenius equation, the results displayed that every NaDELs combination showed Newtonian behavior.[46] Basaiahgari et al.[47] conducted the same study on benzyl tripropylammonium-based DELs and found that data obtained by the VFT model (Equation 6) were more reliable.[47] Likewise, Cui et al.[48] evaluated the viscosity measurements of DELs comprising ChCl as HBA and acidic HBDs. It was observed that the viscosities of DELs fit better with the VFT equation than the Arrhenius model.[48] Another group of researchers[49] presented two different empirical models built from VFT and Eyring viscosity models as a function of composition and temperature and reported that the VFT model was better fitted for DELs with high viscosities.

In contrast, the Eyring model was accurate for low viscosity DELs. For example, the viscosity measurements of DEL combination of levulinic acid and ChCl have fitted accurately with both Arrhenius and VFT models.[50] It has been reported viscosity of a particular DEL is affected by water content,[51] temperature,[52] and the molar ratio of components.[53] In the case of liquid-liquid or solid-phase extraction, an increase in viscosity reduces the mass transfer during liquid-liquid extraction and other chromatographic

separation.[29] Overall, there exists a lack of understanding of the specific origin of viscosity in these liquids and above all, the quantitative effect of DELs viscosity on mass transfer or extraction efficiency needs more systematic investigations.

Density

It has been observed that DELs show higher densities than water (mainly in the range of 1.0–1.5 g cm^{-3}). Basaiahgari et al.[54] calculated the densities of many combinations of HBAs and HBDs and found that glycerol-based DELs had higher densities than DELs based on ethylene glycol (Table 1). A look at the physiochemical data (Table 1) indicates that the formation of hydrogen bonds due to the increase in the number of -OH functional groups signifies the increase in density as it decreases the free available volume.[54] For example, keeping the same ratios between HBA (ChCl) and HBD (organic acid), the DEL comprising oxalic acid was denser (1.30 g cm^{-3}) than malonic acid (1.25 g cm^{-3}). Within the same type of DEL, the highest density of 1.46 g cm^{-3} was exhibited by ChCl:trichloroacetic acid (1:2).[48] It has been seen that the densities of DELs can be adjusted by modifying the molar ratio and overall composition. A plethora of predictive tools have been introduced to anticipate the density of liquids, and many of these models have been successfully applied to estimate the density of DELs. Rackett[55] was the first to modify the equation of state to calculate the molar volume and density (Equation 7) of saturated liquids. However, the model can successfully predict the densities of DELs.

$$\rho = M^{-1}\left\{\left(\frac{RT_c}{P_c}\right)\left(\frac{P_c v_c}{T_c}\right)^{1+\left(1-\frac{T}{T_c}\right)^{0.2857}}\right\} \tag{7}$$

Later on, several amendments in Equation 7 were suggested by Spencer and Danner,[56] Mjalli et al.,[57] and Haghbakhsh et al.[58] The density of DELs in these models was declared to be the function of the critical state variable (P_c, T_c, and V_c) and acentric factor (ω).

A detailed study[59] was conducted by varying citric acid and choline chloride molar ratios. It was observed that an increase in the amount of HBA (ChCl) in the overall combination can decrease the density and vice versa. The densities of DELs can be changed by varying temperatures, where an increase in temperature will decrease the densities. For a particular DEL system, DEL-free volume can be quantified using the isobaric thermal expansion coefficient.[60]

$$a = -\left(\frac{\partial \ln(\rho)}{\partial T}\right))p \tag{8}$$

Yadav et al.[44] reported that densities of glyceline-based DEL showed a linear fit with a quadratic equation in a temperature range of 283 to 363 K. A similar kind of trend (a linear relationship between temperature and DEL's density) exists for nine different combinations of DEL systems.[61] Few other researchers have reported a negative linear relationship with DEL systems like ethylene-glycol-based,[62] choline chloride, varying HBDs,[63] and Ni (II) salt.[64] It has been reported that the density of a particular DEL depends mainly on the molar ratio and the nature of the hydrogen bond donor.[48,53,65]

Surface tension

In addition to density and viscosity, the surface tension of a DEL can play an essential role in its use as an extraction solvent. Besides, the knowledge of surface tension can be linked with the nature of interactions in a particular DEL system. Increasing surface tension can reduce the mass transfer phenomenon during separation processes like extraction, distillation, evaporation, absorption, and crystallization. Likewise, with other physicochemical features, these liquids' surface tension can also be predicted using a quantitative structure-property relationship. The most frequently applied model uses the density data in terms of parachor,[66] or molar volume.[55] The more straightforward ways to calculate the surface tension of liquids involve but are not limited to using a pendant drop,[67] Wilhelmy plate,[68] or a Du Noüy ring.[69] For more comprehensive relationships, we need to correlate the changes in the overall environment of the DEL system and its surface tension.

Gajardo-Parra et al. evaluated the surface tension of three different combinations of DELs systems made up of ChCl and found that ChCl acted as a surfactant to decrease the cohesive forces resulting in a decrease in the surface of the final DEL combinations as compared to the neat components.[63] Hayyan et al. tried to explain the role of the ChCl concentration towards the surface tension its DEL systems. Various combinations of DELs were prepared by varying concentrations of ChCl, and results indicate that increase in the molar amount of ChCl increased the surface tension of DEL.[70] It has been also observed that an increase in temperature results in a linear decrease in the surface tension of a particular DEL.[71,72]

Vapor pressure

The pressure exerted by the vapors of a substance in equilibrium with its condensed phase is termed vapor pressure. The knowledge of vapor pressure is equally important for these designer liquids to work as extraction solvents. Besides, the value of vapor pressure can help us to predict the nature of DEL constituents and the extent of interaction present between these constituents. Nevertheless, very few studies on direct or indirect measurement of the vapor pressure of deep eutectic liquids (DELs) have been conducted so far. The most frequently adopted instrumental approach for the determination of vapor pressure follows Equation 9 of conventional thermogravimetric analysis,[73] in which dm/dt, M, T, and K denote weight loss rate, molecular mass, temperature (K), and constant representing the extent of molecular interaction in a particular DEL system.

$$P^{vap} = k\left(\frac{dm}{dt}\right)\sqrt{\frac{T}{M}} \tag{9}$$

Table 1. A detailed summary of eutectic points and physicochemical properties of various deep eutectic liquids.

HBA	HBD	Ratio	Eutectic point[a]	Viscosity[b]	Density[c]	Surface tension[d]	Conductivity[e]	Refractive index[f]	References
ChCl	Urea	1:2	285	750	1.21	52.00	0.75	1.5044 (303 K)	[51, 105–107]
	Glycerol	1:2	233	246 (303 K)	1.18	57.24	1.74	1.4867	[105, 106, 108, 109]
	Ethylene glycol	1:2	237	48	1.12	48.91	7.61	1.4682	[105, 106, 110]
	Malonic acid	1:1	293	1638	1.25	65.68	0.91	1.4871	[90, 111]
	Oxalic acid	1:1	307	212 (313 K)	1.30		00.48	1.4864	[7, 112, 113]
	Phenol	1:1.5	294	44	1.09	3.14	3.14		[110]
	o-cresol	1:1	249	78	1.07		1.21		[109, 110]
	Xylenol	1:1	290	77					[110, 114]
	trichloroacetic acid	1:2		435	1.46		0.07		[48]
	Acetic acid	1:1		51–162	1.12		3.28		[115]
	Levulinic acid	1:2	366	164	1.14		0.81	1.4679	[7, 116, 117]
	Imidazole	3:7	238	15 (343 K)			12.01 (333 K)		[118]
	1,4-butanediol	1:3	241	140	1.06	47.17	1.64	1.4656	[119, 120]
	D-fructose	1:1	283	183	1.34	70.40		1.5197	[70, 121]
	Glucose	2:1	288	9.03	1.30	71.70		1.6669	[31, 32, 122]
	PG	1:2.5		118 ± 1.08 (323 K)	1.15 (323 K)	39.09 (323 K)	7.16 (323 K)		[123]
	n-butyric acid	1:2	255	1757	0.94				[28, 124]
		1:3	253	54.1					[124]
		1:4	260	13.3					[124]
Benzyltrimethylammonium chloride	EG	1:1		45.3	1.10		4.61		[54]
	Diethylene glycol	1:1		98.6	1.11		1.65		
	Tri ethylene glycol	1:1		308.8	1.12		1.01		
	Glycerol	1:1		753.5	1.18		0.39		
	p-toluene sulfonic acid	3:7		4.861	1.38		2.412	1.689	[125]
	Citric acid	3:7		7.101			4.366	1.552	[125]
	Oxalic acid	3:7		7.178	1.39		5.342	1.691	[125]
Benzyltributylammonium chloride	EG	1:1		286.0	1.04		0.64		[54]
	Diethylene glycol	1:1		310.3	1.05		0.40		
	Tri ethylene glycol	1:1		256.5	1.06		0.27		
	Glycerol	1:1		2328.1	1.10		0.09		
Methyltriphenylphosphonium bromide	Ethylene glycol	1:4		109.8		51.29		1.5622	[52]
	Glycerol	1:3				58.94		1.5666	[52]
	2,2,2-trifluroacetamide	1:8		136.15				1.4849	[52]
Tetrapropylammonium bromide	Ethylene glycol	1:3		58.2 (303 K)		46.20		1.4717 (303 K)	[126, 127]
	Triethylene glycol	1:3		71.9 (303 K)		39.30		1.4836 (303 K)	[118, 126]
	Glycerol	1:4		476.1(303 K)		46.00		1.4858 (303 K)	[118, 127]
	1,3-propanediol	1:3		135 (303 K)					[128]
	imidazole	3:7		314.5 (303 K)					[118]
	1 1,5-propanediol	1:3		183 (303 K)					[128]
N, *N*-dimethylammonium chloride	Ethylene glycol	1:2		40.68 (303 K)		47.51		1.4676 (303 K)	[129]
	Glycerol	1:2		351.46 (303 K)		59.35		1.4843 (303 K)	[129]

(*continued*)

Table 1. Continued.

HBA	HBD	Ratio	Eutectic point[a]	Viscosity[b]	Density[c]	Surface tension[d]	Conductivity[e]	Refractive index[f]	References
NaCl	Ethylene glycol	0.5–20		20.883 (303 K)	1.14 (303 K)		4.73 (303 K)	1.4353 (303 K)	[130]
NaBr	Ethylene glycol	0.5–10		39.03 (303 K)	1.22 (303 K)		6.92 (303 K)	1.4546 (303 K)	[130]
NaI	Ethylene glycol	0.5–10		47.72 (303 K)	1.48 (303 K)		7.81 (303 K)	1.4873 (303 K)	[130]
$ZnCl_2$	Urea	1:3.5		11,340	1.63		0.18 (315 K)		[131]
	Acetamide	1:4			1.36				[131]
	1,2-ethanediol	1:4			1.45				[131]
	1,6-hexanediol	1:3			1.38				[131]
	Glucose	1:1			1.00		0.06		[132]
	Fructose	1:1			1.12		0.01		[132]
	Lactic acid	1:1			1.32		0.01		[132]
Water			273	0.89	0.99	71.97	0.05	1.333	[133]
Ethanol			159	1.07	0.78	21.97	0.55	1.361	[133]
Ethyl acetate			190	0.423	0.90	24.01	0.03	1.372	[133]
Diethyl ether			156	0.244	0.71	17.06	0.13	1.355	[134]

The values are rounded off for coherency and units are a = Kelvin at 1 atm pressure, b = centipoise (cP) at 298 K, c = g/cm^3 at 298 K, d = mN m^{-1} at 298 K, e = mS cm^{-1}, and f = n_D at 298 K or stated otherwise.

Boisset et al.[74] calculated the vapor pressure of the DELs system comprising lithium bis-imide and N-methyl acetamide at 313 K in a 1:4 molar ratio to be 20 Pascal which is very low as compared to water.[74] Mjalli et al.[57] also applied the above equation to determine the vapor pressure of five different DELs combinations using and reported their values to be the same as those of Boisset et al.[75] Ravula et al.[76] found that di-cationic ionic liquids had the lowest vapor pressure values while DELs possessed only lower than short and long-chain polyethylene glycol. Dietz et al.[77] determined the vapor pressure of six different DELs combinations by headspace gas chromatography-mass spectrometry (HS-GC-MS) and found that the vapor pressure of many DEL systems is low compared to organic solvents. Recently, Dietz et al.[78] applied perturbed-chain statistical association fluid theory (PC-SAFT) to determine the total vapor pressure of hydrophobic DEL systems. The PC-SAFT modeling provides an opportunity to assess the activity coefficients of HBDs and HBAs of DEL. Besides, this method can predict the total vapor pressure of liquid from its constituents' vapor pressure data, which implies significant time and cost-saving.

Miscellaneous

The refractive index (RI) is a dimensionless property that compares light speed when it passes through space and DEL.[79] Seki et al.[80] prepared different molar combinations of DEL comprising ChCl and glycerol and found that eutectic composition had the highest value of RI. A study conducted by Su et al.[81] describes the effect of the type and concentration of HBD on the refractive index of DEL keeping HBA same. The results indicate that DEL containing tetrabutylammonium chloride (TBAC) and phenylacetic acid had a greater RI than DEL containing TBAC and propionic acid due to the bulky phenyl group. Similarly, Sánchez et al.[82] evaluated the molar refractivity of seven DEL combinations of L-proline and betaine and found that it depends on HBD molar concentration.

In addition to the above-cited features, the pH of a DEL system is a vital factor for defining its characteristics along with other industrial applications. Hayyan et al.[31] tested the pH of various DEL combinations of varying molar compositions of ChCl and glucose. They reported a linear dependence of pH with temperatures from 25 to 85 °C and concluded that these mixtures could be used commercially.[31] Skulcova et al.[83] reported a linear decrease of pH as the temperature was increased for seven different combinations of DEL with significant influence created by the type of HBD used.[83] Meanwhile, a few authors like Zhang et al.[84] compared the extraction efficiency of different combinations of DEL by varying pH from 4 to 10 and found that pH had no significant effect on the extraction efficiency of proteins.

Due to their high ionic conductivity, DEL finds potential uses in power systems like advanced electrolytes for batteries.[85–87] It has been reported that the ionic conductivity of these solvents is lower than salts in molten states. According to hole theory, Abbott et al. explained this difference based on ionic size and viscosity.[88] In another work, Abbott et al. reported that the ionic conductivity of DEL could be increased by decreasing the size of the cation, which will increase the free volume of DEL, resulting in decreased viscosity and increased ionic conductivity.[89] Overall, ionic conductivity depends on donor-to-acceptor ratios and water content in a particular DEL.[90,91]

Effect of dilution on physicochemical features of DELs

It has been noticed that deep eutectic liquids (DELs) absorb water from the air due to their hygroscopic nature, which influences the properties of the solvent. Although water absorption during preparation may work as an impurity, some researchers intentionally add water in DELs to tweak its properties for the desired application. The reports have suggested that adding water to a particular DEL leads to changes in various physicochemical properties, mainly density, melting point, surface tension, viscosity, polarity, and conductivity. Meng et al.[92] evaluated the change in the melting point of DEL containing ChCl and urea by adding water. The solvent's melting point decreased from 30 °C to 15 °C by adding 5% water.[92] Another group also evaluated the effect of dilution on the physiochemical properties of DEL-based on ChCl and urea. These researchers observed a remarkable decrease of up to 22% in viscosity and a slight reduction of 0.14% in density by adding a 0.008 mass fraction of water.[93] In another report, adding 6% water in ChCl and urea-based DEL increased conductivity by ten times and decreased viscosity by 16 times.[94] Paradiso et al.[95] found that adding water into a DEL comprising glucose and lactic acid decreased the solvent's viscosity to a significant level where it can be used for various extractions.[95] Shah and Mjalli[51] observed that conductivity of ChCl and urea-containing DELs can increase up to three folds for 10% dilution with water which was associated with an 80% decrease in viscosity. A similar kind of behavior was observed for the density, viscosity, and conductivity of various NaDELs.[91] Gabriele et al.[96] studied the impact of water content on conductivity and viscosity of DELs comprising ChCl as HBA and various HBD species and found that viscosity of a particular DEL was almost halved at 10% water content while the conductivity increased by 15 times. The highest value of conductivity was observed for the DELs containing 60% water. Nunes et al.[71] investigated the effect of water content on the surface tension of a DEL composed of octanoic acid and menthol and observed that an increase in water content initially decreased the surface tension, followed by an increase. A similar trend has been also observed for DEL containing malonic acid and ChCl.[97] It can be speculated that water in a particular DEL affects the network of hydrogen bonds and other interactions. It should be further kept in mind that water itself can act as HBD to strengthen the binding forces up to specific water content, after which dilution weakens the overall network's strength.[15,98,99] However, in addition to the alteration of

properties, the integrity of a DEL system can also be jeopardized by the inclusion of water.[100]

An increase in the number of bonding network arrangments makes these liquids complex to study and predicted but raises an opportunity of tunable character for end-user applications. The researchers have found that most of the interactions in these liquids are intramolecular bonds between HBA and HBD. At the same time, even complex structures are possible due to the establishment of hydrogen bonds between HBDs and HBAs. During molecular studies, four types of H-bonds have been located whose relative contribution depends upon the mixing ratio of HBD and HBA. In the case of ChCl and alcohols, the HBA possesses a spatial location near HBD.[101]

Meanwhile, it has been speculated that deep eutectic liquid keeps its structure for up to 50% water content but thermophysical properties like viscosity, surface tension, and density decrease with the increase in water content. The radial distribution function (RDF) has also been applied to probe the variation in structural features of deep eutectic mixtures with different water contents. The RDF discloses the formation of isosbestic-like points for chloride-water and chloride-urea interactions whose proportion (but not arrangement) increases with the increase in hydration level.[102] In the case of hydrophilic deep eutectic liquids like ChCl/Urea or ChCl/ethylene glycol, the presence of water even below traces may form a quaternary system in which water can act as HBD or HBA depending upon its molecular environment. The experimental and theoretical modulation indicates that water can compete with any of constituting HBD or HBA and disrupt its ratio, thus making the mixture a quaternary system. According to Vilková et al.,[102] the absence of isosbestic-like points for chloride-chloride and chloride-choline interactions suggests that two types of structures exist. Several studies have already claimed heterogeneity or the presence of microphases, *i.e.*, *water-in-deep eutectic mixture* or *deep-eutectic mixture-in-water*. The change in hydration level mediates transitions between these microphases.

In the case of hydrophobic DELs, Kivelä et al.[103] comprehensively studied the structural variation and change in the thermophysical characteristics of a hydrophobic deep eutectic liquid (HDEL) comprising a 1:2 mixture of decanoic acid and tetrabutylammonium chloride. The presence of water even below 2.5% (wt) can significantly affect the physicochemical features of the mixture formed. In particular, the density, viscosity, and melting point of HDEL increase linearly with the increase in water content but the electrical conductivity exhibit a nonlinear dependence because of changes in the charge carriers. The molecular dynamics and simulations study indicate the structural clustering involving hydration on the chloride-carboxylic acid hydrogen bonds. In the absence of water, chloride binds to decanoic acid via hydrogen bond and then forms ion pairing with tetrabutylammonium chloride's strong electrostatic interaction. The intermolecular interaction produces a molecular network in which molecular motions become restricted. The presence of water may break the interaction between chloride and chloride-carboxylic acid and induce the ion pair formation between tetrabutylammonium and chlorides. Water can also hydrate carboxylic acids and chlorides to induce dynamic nanoscale phase segregation. The authors also claim that the composition range for the liquid solvent becomes wide due to the higher hydrogen bonding capacity of chlorides. Rodriguez et al.[104] also noted that water in the combination of ChCl and carboxylic acids could avoid esterification. However, the esterification reaction may accelerate during drying. The authors recommend that ChCl and carboxylic acid type binary mixtures can only be applied once the moisture level or ester formed is adequately quantified. Zhekenov et al.[15] have noted that adding water in a particular DEL improves the interaction between water and HBD, cation of HBA and water, and anion of HBA and water. In the case of ChCl:U and ChCl:Glycerol, the strength of the hydrogen bond between chloride-HBD decreases drastically. Likewise, the addition of water up to 5% (w/w) can get absorbed in the molecular matrix of liquid and does not affect the physicochemical attributes of liquids. However, the further addition of water will dampen the intermolecular and intra-molecular interactions drastically.

Green features of deep eutectic liquids

With the emergence of the green chemistry approach, the exploitation of eco-friendly and benign liquids has become indispensable. For the last couple of decades, deep eutectic liquids (DELs) particularly those composed of natural HBD and HBA species have been considered green and eco-friendly for certain traits. First of all, the intercalation between HBD and HBA species can produce liquids practical nonvolatile, and benign to the environment. The type III and advanced types of DELs comprising tertiary amines, carboxylic acids, and alcohols are often biodegradable with no toxicity, and low reactivity with water makes them an ideal solvent for various analytical applications. Besides, the low- production cost with a great variety of readily available raw materials is the cherry on top.[5,135]

The tailorable features of DELs have recently attracted researchers heading toward green chemistry routes. Likewise, the variation in the mobility, density, viscosity, surface tension, and electrical conductance of DELs with the change in nature/mixing ratios of HBD/HBA and moisture level/water content can also work as a new areas of research. The researchers now have adopted theoretical and practical models to explore the effect of water on the physicochemical characteristics of DELs. As many of the DEL formulations contain compounds with low or zero toxicity (including food-grade components like glucose, fructose, and certain amino acids), they have the potential to be evaluated as extraction and synthesis solvents. Many studies have been conducted in this regard to grade the cytotoxic profile of DELs. Haerens et al.[136] reported that DELs comprising ChCl and EG were harmless to the environment compared to most ionic liquids, which possess a certain threat. These authors also found above DELs recyclable from various materials but further studies are required to establish this

opportunity.[136] Nevertheless, it would not be fair to declare the DELs generally regarded as safe (GRAS) with the available record of clinical data but those containing natural HBD and HBA species like sugars, organic acids, and certain amino acids can be declared nontoxic or eco-friendly as the components are biodegradable and nontoxic.[137,138] Huang et al.[139] prepared thirteen different combinations of NaDELs using ChCl, Glycerol, sugars, amino acids, and claimed that the majority of prepared NaDELs were nontoxic and biodegradable. Likewise, Bajkacz and Adamek[140] claim that NaDELs comprising ChCl: Glucose, Lactic acid: Glucose, and Fructose: Glucose: sucrose lactic acid: glucose, glucose: choline chloride, and fructose: glucose: sucrose are nontoxic and biodegradable.

Recycling of DELs

Reusing/Recycling solid catalysts and liquids solvents have been a highly appreciated and desirable approach for sustainability and economic point of view. Nevertheless, it is always difficult to recycle a good solvent for thermodynamic constraints during isolation, enrichment, and/or purification. Needless to say, these steps often become necessary to isolate the solutes/analytes of interest while using DELs as an extraction solvent. There exist a lack of sound knowledge and understanding about energetics and mechanisms to recycle a DEL combination after use, still, many researchers undertaking extraction of plant bioactives from agricultural resources or spent electrode materials from lithium-ion batteries claimed these liquids are re-useable up to four to five cycles.[14,141] Some researchers also observed a decline in the extraction efficiency, performance, and overall composition of the tested DEL System.[142,143] For example, Di Gioia et al.,[144] claim no loss of performance for a DEL system comprising ChCl and urea (1:2) after four cycles of application, while Kumar et al.[107] found the NaDELs comprising lactic acid and ChCl (5:1) was effective for lignin extraction till four cycles. Another study says that xylan solubility in the aqueous solution of ChCl and acetic acid-based DEL decreased by 5% after the first two cycles, while DEL constituents could be recycled up to four times.[145] The ways followed for recycling a particular DEL solvent involve evaporation under reduced pressure.[99,146–148] No doubt few researchers checked the purity of recycled DELs by monitoring particular functional groups via FT-IR.[149] However, these methods are only effective to concentrate a DELs system via evaporation of water or other volatiles. The DELs have very low vapor pressure and likewise, the analytes with low vapor cannot be evaporated rotary type evaporators. The pervaporation or reverse osmosis may work as an alternative choice for the recycling of DELs but this technology also fails for higher moisture levels and ionic strength. The magnetic field can also be a good opportunity to recycle DEL after application, this approach has been applied to recover DEL composed of ChCl and tetra-cholorferrate ($FeCl_4$).[150]

The recycling of DELs from extraction or reaction media needs more exhaustive and comprehensive studies to derive through a straight-way methodology and right now it would be fair to say that recycling or isolation of deep eutectic liquids after their application is still an un-addressed challenge regarding the utilization of these green liquids as solvents.

Applications of deep eutectic liquids (DELs) as extraction solvents

The advent of modern chromatography and spectroscopic instruments has revolutionized the resolution and sensitivity of analytical assays. Meanwhile, it has been realized that the results of the chromatographic or spectroscopic analysis can be as good as the sample preparation or extraction step had been. The later theory diverted the whole community of analytical chemists to work on some robust and efficient sample preparation or extraction methods. The use of toxic and volatile organic solvents for the dissolution/extraction/elution/washing of the samples or standards has been extensively criticized from human health and environmental safety point of view. The discovery of DELs has opened new hope in many areas of life and physical sciences, particularly in the under-addressed areas of sample preparation or extraction. During the last couple of decades, the DELs have been extensively investigated for the extraction or purification of analytes of almost all types (organic, inorganic, proteinic, volatile, nonvolatile, etc.).

The data plotted in Figure 4a and b manifests the steadily growing trend in the research published on DELs (b) and their use as an extraction solvent (a). The earlier trend of DELs usage for various extraction procedures started with type III DEL consisting of ChCl as HBA paired with glycerol, urea, or ethylene glycol as HBDs.[151] However, with the advent of time, alternative HBDs like amides, alcohols, sugars, amines, carboxylic acid, and HBAs such as imidazolium salts, amino acids, sugars, tetraalkylammonium salts, and carboxylic acids got promoted.[152] Hydrophobic DELs are commonly used as solvent systems in aqueous samples for better phase separation than traditional solvents.[153]

It has been noted that the properties of DELs are somewhat different from the individual components. These liquids, unlike the organic solvent, could not be applied in pure, instead, the researchers need to incorporate these liquids with conventional solvents like water, ethanol, or methanol for liquid-liquid extraction (LLE), heat-assisted extraction (HAE), microwave-assisted extraction (MAE), or ultrasound-assisted extraction (UAE) as manifested from the generic layout shown in Figure 5. The nature of co-solvent, types of irradiation (thermal, microwave, or ultrasound), the mechanism of separation (partition or dispersion), and application scale decide the type of extraction technique.

These assumptions can be justified by the literature reports summarized in Table 2. Li and Row[154] reviewed the application of DELs in liquid-liquid microextraction and declared these liquids as designer solvents for the extraction of biomolecules. Meanwhile, Makoś et al.[155] also highlighted the potential of deep eutectic liquids in microextraction. Van Osch et al.,[156] and Dwamena[157] pointed out the

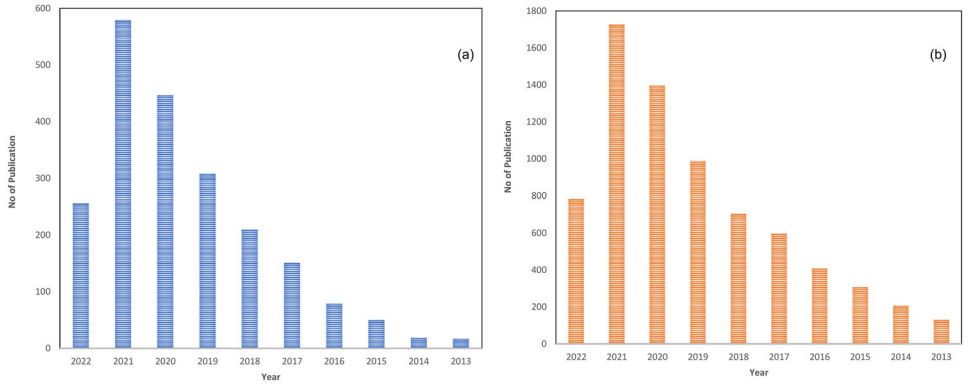


Figure 4. A Comparison of the relentless increase in the number of research reports for DELs (b) and their application as extraction solvent (a).

features and potentials of hydrophobic deep eutectic liquids in extraction and separation science. Recently, Lee et al.[158] also recommended hydrophobic DELs for the extraction of organic and inorganic analytes. Cunha and Fernandes[14] reviewed the opportunities for ultrasound and microwaves during the DEL-assisted extractions.

It should be mentioned here that the low melting points and vapor pressure of DELs have made these liquids green and sustainable but the resultant increases in viscosity, density, and surface tension have challenged their applications in solvent-based extractions. Apart from that, the opportunity of changing these properties by tweaking the composition has motivated all the researchers across the map to test DELs as extraction solvents for polar and non-polar compounds, including phenolic compounds and flavonoids, drugs, toxins, heavy metals, and insecticides.[159] In majority of these compounds the target analytes have been quantified by High-Performance Liquid Chromatography coupled with Diode array Detector (HPLC-DAD), Atomic Absorption Spectroscopy (AAS), Gas Chromatography (GC) coupled with flame ionization detector (FID), Electron Capture Detector (ECD), or Electrospray Ionization type Mass Spectrometry (ESI-MS), or simply ultra-violet visible spectroscopy (UV). This section describes some typical examples regarding the utilization of DELs in different types of extraction methods.

Liquid-liquid extraction (LLE)

The foods derived from different plant species are reported to contain numerous bioactives, which are of prime importance due to their potential biological activities. In addition to applying DELs in various fields of life and physical science, these liquids have been relentlessly applied to extract bioactive compounds from agricultural sources by applying LLE. For example, Đorđević et al.[160] applied ChCl:urea (1:2) and triethanolamine:glycerol (1:2) like DELs, diluted (75/25 v/v) with ethanol and water for the extraction of phenolic compounds from black mustard seed. The authors found that mixing triethanolamine:glycerol-based DEL with ethanol (75/25 v/v) offered higher recovery of phenolics like ellagic, syringic, gallic, and caffeic acids from the powder mustard black seeds. Likewise, Wojeicchowski et al.[161] found that the DEL system comprising ChCl as HBA and 1,2-propanediol as HBD premixed with water (50:50 v/v) when applied at 65 °C at a liquid to solid ratio of 40 recovered 220% more phenolics. de Almeida Pontes et al.[162] claimed 15% higher recovery rates of phenolics from olive leaves while applying ChCl diluted with water (50:50 v/v) at 54 °C and 400 rpm of shaking speed. Liu et al.[163] tested the efficacy of different combinations of DELs for the extraction of tocopherols from soybean oil. The results showed that DEL containing *p*-cresol and ChCl exhibited the best extraction efficiency for all required isomers. This DEL combination may form strong π-π interaction with tocopherols. Abdul Hadi et al.[109] also evaluated the efficacy of DELs based on ChCl and carboxylic acids for the extraction of tocols (tocopherols and tocotrienols) via liquid-liquid extraction. It was reported that the use of DELs promoted the recovery of tocotrienols and overall yield increased with increasing the amount of DEL (Table 2). Aydin et al.[114] were able to extract and purify curcumin from various food and tea samples via vortex-assisted extraction using DEL containing ChCl and phenol in combination with organic solvent tetrahydrofuran, which acted as an emulsifier. The spectroscopic analysis showed a high concentration of curcumin in the extract obtained at pH 4.0 with this DEL combination.[114]

Tan et al.[164] used various DEL combinations to extract plant growth regulators from different vegetable oils and found that a DEL made up of ethylene glycol and tetraethylammonium chloride offered good extraction efficiency for three plant growth regulators.[164] Qi et al.[165] prepared nine combinations of DEL to extract flavonoids from *Equisetum palustre*. Their results showed that the polar flavonoids were extracted with the highest efficiency in DEL based on choline chloride, betaine, and ethylene glycol. Nam et al.[166] tested the efficacy of various DEL combinations for extracting flavonoids from *Flos sophorae*. The best extraction efficiency of quercetin was noted with DEL consisting of choline chloride and glucose, while choline chloride and xylose were best for the extraction of kaempferol and isorhamnetin. Tang et al.[167] utilized different DEL combinations to extract Myricetin from *Chamaecyparis obtusa* and found that the targeted flavonoids were extracted with maximum efficiency in DEL based on choline chloride and 1,4-butanediol.

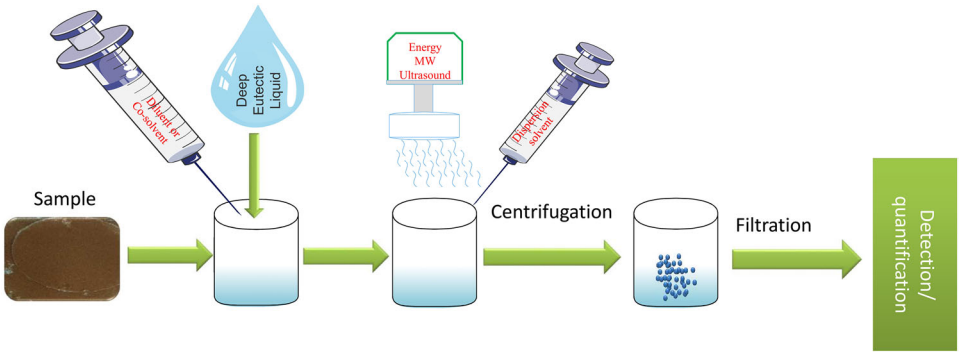


Figure 5. The typical layout adopted for various types of extraction techniques (LLE, UAE, MAE, and DLLME) while applying deep-eutectic liquids.

The natural deep eutectic liquids (NaDELs) have been considered to be more green and benign solvents for the extraction of flavonoids from various natural products.[138,168] A series of flavonoids, including chrysin, naringenin, rutin, quercetin, hesperidin, and neo hesperidin, was extracted from various fruit and vegetable samples using different combinations of NaDELs as extraction solvents (Table 2). A NaDEL combination containing lactic acid and acetylcholine chloride (1:2) when mixed with 30% (v/v) water (for decreasing viscosity of DEL) exhibited higher extraction yields of flavonoids containing hesperidin, rutin, neo-hesperidin, naringin, naringenin, quercetin, chrysin, and hesperetin. Guo et al.[108] Guo et al. extracted anthocyanins from fresh mulberry using different combinations of NaDELs paired with homogenization at high speed. The best extraction efficiency was obtained from NaDELs based on choline chloride, glucose, and citric acid (1:1:1), with significantly higher values than NaDELs based on alcohols and conventional solvents.

In addition to polyphenolic compounds and flavonoids, various DEL combinations have been employed to extract and quantify other polar compounds from different samples. One of the most famous synthetic dyes in the chemical industry in Sudan is forbidden in the food industry due to genotoxicity. Liu et al.[169] developed a vortex-assisted extraction system based on NaDELs to identify the presence of Sudan I in various food samples. The study showed that NaDELs based on choline chloride and sesamol in a molar ratio of 1:3 displayed excellent extraction efficiency of Sudan I due to its π-π interaction and hydrogen bonding with the targeted compound.

Due to its various health benefits and being a potential candidate to substitute alcohols, the extraction and enrichment of propolis are of great importance to the food industry. Funari et al.[170] developed a novel method utilizing sixteen different combinations of green biodegradable NADELs to extract propolis without undertaking the recovery of propolis from NaDELs as the solvent is biocompatible and nontoxic with excellent stability over months. Liu et al.[171] utilized some novel DELs to extract protein from various plant and food sources and found that ChCl and Polyethylene glycol (PEG-200) when combined with microwave and ultrasound irradiation successfully precipitated protein from the solution via co-precipitation method. Due to its numerous uses in the food industry, the extraction and enrichment of gluten are of prime importance. Svigelj et al.[172] developed a method to extract gluten from processed and unprocessed food samples using DEL based on choline chloride. The results showed that DEL based on choline chloride and urea or ethylene glycol yields 60% more effective than conventional solvents. In the same way, the protein was extracted from various oilseed cakes like rapeseed using DEL based on choline chloride and glycerol. It was found that protein precipitates can be obtained using water in the solution containing extract, which will act as an anti-solvent at 60 °C.[173]

Dispersive liquid-liquid microextraction (DLLME)

Mogaddam et al.[174] applied dispersive liquid-liquid microextraction (DLLME) for the extraction of acaricides containing fipronil sulfone, amitraz, cypermethrin, and carbaryl from hen egg using NaDELs. The authors applied water miscible DEL (ChCl:ascorbic acid) along with zinc acetate and NaCl to salt out acaricides which were subsequently dispersed in water and ChCl -acetic acid-n-octanol DEL. The DLLME using NADELs offered a simple, rapid, green, and in-expensive solution for the determination of acaricides. Afshar Mogaddam et al.[28] also applied *in-situ* prepared ChCl:*n*-butyric acid based DEL for dispersive solid and liquid phase microextraction (DSPE and DLLME) of Phytosterols (stigmasterol, campesterol, brassicasterol, lupeol, and β-sitosterol) from vegetable oils (sunflower, hazelnut, canola, corn, walnut, almond, olive, grape). Mardani et al.[175] has successfully dispersed organochlorines (aldrin and dichlobenil) salted out from cocoa beans into diethanol ammonium chloride (DEAC) and pivalic acid-based DEL. A new version of DLLME i.e., DEL-based evaporation assisted DLLME was developed by Nemati et al.[176] that offered an enrichment factor between 623 and 690 for the extraction of pyrethroid from fruit juices. In another variant of DLLME, ultrasound-assisted emulsification liquid phase microextraction (UA-LPME) was introduced by Aydin et al.,[177] and his coworkers[178–180] to extract malachite green, chromium, cobalt, and tartrazine from fish farming water, tea, and drugs, respectively (Table 2). The key advantages attributed to UA-ELPME include their simplicity, rapidness, and green credential.

Wang et al.[181] introduced a liquid-liquid micro-extraction (LLME) method for efficiently extracting and

Table 2. Application of deep eutectic liquids (DELs) as an extraction solvent for diveristy of analytes.

Sample/Matrix tested	DEL system	Co-solvent (the liquid mixed with DEL system)	Detection/Quantification	Major constituents	Extraction efficiency/Yield	Reference
Liquid-liquid extraction (Agitation 100–250 rpm) at 40–80 °C for 6–12 hours						
Celery seed	Betaine HCl:PG 1:8 + 40% HOH	40% (w/w) water, water acted as anti-solvent	HPLC-DAD	apigenin and luteolin	516.58 and 444.59% higher enrichment factor	[29]
Juglans regia L.	ChCl:Butyric acid	20% (w/w) Water	HPLC-DAD	Phenolic compounds	32.5 mg/g	[206]
Rice straw biomass	Lactic acid:ChCl (1:5)	Water (5–50%)	UV	Lignin	68–72 mg lignin/g of biomass	[107]
Black mustard seeds	ChCl:U 1:2 Triethanolamine:Glycerol 1:2	H_2O (75/25 v/v) or C_2H_5OH (75/25 v/v)	HPLC-DAD	Ellagic, syringic, gallic, and caffeic acid,	32.0–32.5 mg/g for Triethanolamine:Glycerol 1:2 + C_2H_5OH	[160]
Rosemary leaves	ChCl: 1,2-propanediol	Water (50:50 v/v)	HPLC-DAD	Phenolic compounds	78 mg/g	[161]
Chamaecyparis obtusa	methyl triphenyl phosphonium bromide:ethylene glycol (1:5)	Methanol (50:50 v/v)+ heating at 60 °C for 120 min at a solid-liquid ratio of 80%	HPLC-DAD	Myricetin, quercetin, and amentoflavone	325.90 μg/mL of quercetin, 111–128% increase in extraction Yiled	[167]
Olive leaves	ChCl:Acetic acid	Water (50:50 v/v)	UHPLC-DAD	Kaempferol, ferulic acid, oleuropein, tyrosol, and luteolin	15% (TPC 470 mg/Kg) more phenolics than conventional solvent	[162]
Sophora japonica	ChCl:1,4-butanediol 1:4	20% water (v/v)	HPLC-DAD	Rutin	194.17 ± 2.31 mg·g^{-1}	[207]
Fruits and vegetables	acetylcholine chloride:lactic acid 1:2	30% water (v/v)	HPLC-DAD	hesperidin, rutin, naringin, naringenin, quercetin, and chrysin,	≥70% Yield	[208]
Crude palm oil	ChCl:Malonic acid 1:1	Methanol	HPLC-DAD	tocotrienols	Extraction yield (8671 mg/kg)	[109]
Myrothamnus flabellifolia	Sugar-based NaDELs	25% ultrapure water	HPLC-DAD	Phenolics	–	[209]
Fish	ChCl:Oxalic Acid (1:2)	DEL assisted digestion of the fish sample	AAS	Selenium and Arsenic	99.6 and 99.3%, respectively	[210]
Environmental samples	ChCl:urea 1:1.23	Mixed with Fe_3O_4 nanoparticles	AAS	Lead and Cadmium	95–105%	[205]
Dispersive Liquid-Liquid Microextraction (DLLME)						
Hen egg samples	ChCl:Ascorbic acid 1:1.25 & ChCl:AcOH:*n*-octanol DES	In the first phase, acaricides were dispersed in the water-miscible DEL, then water-immiscible	GC-FID	acaricides like fipronil sulfone, amitraz, cypermethrin, and carbaryl	79–92%	[174]
Edible oil Samples	ChCl:Butyric acid	Dispersed analytes were isolated by freezing the organic layer of DLLME	GC-MS	Phytosterols	82–97%	[28]
Cocoa been samples	DEAC and pivalic acid	Salted-out pesticides were dispersed by DEL and Organic solvent	GC-ECD	Organochlorines aldrin and dichlobenil	74–92%	[175]
Fruit juices	ChCl:butyric acid	The organic after mixing with DEL was evaporated	GC-MS	pyrethroid pesticides	83–92%	[176]
Fish farming water	ChCl:Phenol (1:4)	Ultrasound-assisted emulsification followed by extraction with DEL	UV	Malachite green	95%	[177]
Pharmaceutical supplements and tea	ChCl:Phenol (1:4)	UA-ELPME	AAS	Cobalt	97–105%	[178]
Beverage, water, and drugs	Tetrabutylammonium chloride:Decanoic acid	UA-ELPME	UV	Tartrazine	≥90%	[179]
Microwave-assisted extraction (MAE)						
Brewer's spent grain	ChCl:glycerol	Water (75:25 v/v)	HPLC-DAD	Phenolic compounds	2.89 mg/g	[211]
Olive leaves	ChCl:ethylene glycol	Water	LC-MS	Phenolics	29.49 mg/g	[212]
Olive oil wastes	ChCl:glycerol	Water	HPLC-DAD	Phenolics	8.31 ppm	[213]
Mulberry leaf	ChCl:glycerol	Water (80:20 v/v)	HPLC-DAD	Phenolics	7.642 mg/g	[214]

Lippia citriodora	ChCl:Lactic acid	Water (75:25 v/v)		Phenolics	73.13 mg/g	[215]
Lonicerae japonicae	ChCl: 1,3-butanediol	Water	HPLC-DAD	Chlorogenic acid	26.07 mg/g	[216]
Cajanus cajan leaves	ChCl:Maltose	Water (0%, 20%, 40%, and 60%)	UHPLC-DAD	Phenolic acids	30.63 mg/g	[217]
Pigeon pea roots	ChCl:1,6-hexanediol 1:7	30% of water in DEL, 600 w	HPLC-DAD	genistin, genistein and apigenin	Yield can reach 0.449, 0.617 and 0.221 mg/g DW, respectively	[218]
Tuna fish	ChCl:hexanedioic acid	microwaved and mixed with different amounts of HBD and HBA	HPLC-DAD	Biogenic amines	82–93%	[197]
Ultrasound-assisted extraction (UAE)						
Carya cathayensis Sarg	ChCl:Malic acid	Water	HPLC-DAD	Phenolic compounds	49.70 mg g	[219]
coffee and cocoa	ChCl:Lactic acid	10% (v/v) water	HPLC-DAD	Phenolic compounds	44.2 ± 1.09 g/100 g	[220]
Rosmarinus officinalis L.	ChCl: 1,2-propanediol	10% (w/w) water	HPLC-DAD	Phenolic compounds	62.21 ± 3.85 mg/g	[221]
Extra-virgin olive oil	Betaine:glycerol	Water (1:1, 2:1, 3:1, w/v)	HPLC-DAD/ESI-MS	Phenolic compounds	773.03 ± 15.11 μg/g	[222]
Moringa oleifera L.	Alcohol-based DEL	Water (70:30 v/v)	LC-MS	Phenolic compounds	32.6 ± 7.2 mg/g	[223]
Chinese dark tea	ChCl:Lactic acid	70 wt% DELs aqueous solution.	HPLC-DAD	Phenolic compounds	26.78 mg/g	[224]
Morus alba L.	ChCl:Citric acid	Water (3:1 v/v)	HPLC-DAD	Phenolic compounds	22.66 mg/g	[225]
Ficus carica L	Glycerol:Xylitol:Fructose	Water (70:30 v/v)	HPLC-DAD	Bioactive compounds	15.22 mg/g Psoralen and 5.207 mg/g Rutin	[112]
Grapes	ChCl:Malic acid	Water (70:30 v/v)	HPLC-DAD	Phenolics	91mg/g	[226]
Grape skin	ChCl:Oxalic acid	10%, 25% and 50% w/w (Water)	HPLC-DAD	Phenolic compounds	NA	[196]
Mint	Lactic acid:Sodium acetate	80% v/v DELs aqueous solution.	UV	Total Polyphenols	108.24 mg/g	[227]
Prunella vulgaris	ChCl:Ethylene glycol	Water (0–100 vol%)	UHPLC-DAD	Rosmarinic acid	3.658 mg/g	[228]
Gardenia jasminoides Ellis	ChCl:PG 1:2.5 + 35% HOH	35% (v/v) water	UHPLC-MS	crocins	36.97 mg g^{-1} DW	[123]
Cicer arietinum L.	ChCl:propylene glycol (1:2)	10, 30, and 50% water (v/v)	UHPLC-MS	Isoflavones	8.35 mg/g	[229]
Lycium barbarum L.	ChCl:*p*-toluene sulfonic acid	5, 15, 25, and 40% water (v/v)	HPLC-DAD	Flavonoids	57.2 mg/g myricetin and 9.1 mg/g rutin	[230]
Tartary buckwheat hulls	ChCl:glycerol (1:1)	0–50% (w/w) water	HPLC-DAD	Rutin	9.5 mg/g	[142]
soy products	ChCl:Citric acid (1:1)	30% water	UHPLC-DAD	Isoflavones (daidzin, genistin)	64.7–99.2%	[140]
Mint	ChCl:Lactic acid	80% aqueous solution of DEL	UV	Total flavonoids	24.89 mg/g	[227]
Chinese medicine (*Flos sophorae*)	Glycerol:l-proline 1:5	10% water (w/w)	HPLC-DAD	Flavonoids	81–87%	[166]
Mouse Liver Protein	ChCl:U 1:2	Elution from Molecularly imprinted DEL on monolith by $CH_3COOH/ACN/H_2O$ 5/5/90 (v/v)	LC-MS/MS	Acetylated Proteins and peptides	3.1–3.9 folds higher	[198]
Brown sugar	TEAC:EG 1:2	Eluted with 10 mL *n*-hexane and 5.54 mL acetonitrile from Impregnated Silica Gel.	HPLC-DAD	triazine herbicide	95.0–101.7%	[199]
Rice, millet, wheat, and barley	L-proline:EG	Biomass carbon quantum dots (CQDs) were prepared and monitored at 370 nm	Fluorescence detector	carbamate insecticide Methomyl	86.5–107.8%	[200]
Water and food	Menthol:lauric acid 1:3	Sample in aqueous phase was mixed with DEL, sonicated, centrifuged, and analyzed AAS	AAS	Selenium	91.4–103.1	[231]

quantifying four herbicides common in edible oils using DEL. They found that DEL based on ethylene glycol and tetrabutylammonium chloride showed the best vortex-assisted extraction of the targeted compounds. Torbati et al.[182] Torbati et al. introduced a method based on air-assisted LLME using DEL coupled with Gas Chromatography-Mass Spectrometry (GC-MS) for chemical characterization of various aromatic amines in water.

Ultrasound-assisted extraction (UAE)

Huang et al.[123] applied ChCl, and PEG (1:2.5) based deep eutectic liquid along with 35% (v/v) of water as an extraction solvent for ultrasound-assisted extraction (UAE) of crocins from (*Gardenia jasminoides* Ellis). The solvent system comprising DEL of ChCl and PEG was labeled as tailor-made and suitable for affinity-based crocins extraction. The prepared solvent system offered higher surface tension, polarity, and electrical conductivity. However, its extraction efficiency was attributed to the extent of available hydrogen bonds. Liu et al.[113] also evaluated the efficacy of various DEL combinations for efficient extraction of sesamol from sesame oil via ultrasound-assisted liquid-liquid microextraction (UALLME) which was subsequently determined by HPLC-DAD analysis. The authors claim that DEL, composed of ethylene glycol and ChCl offered a better yield regarding the extraction of sesamol.[113] Khezeli et al.[183] also employed the UALLME technique to extract three phenolics (cinnamic acid, ferulic acid, and caffeic acid) from cinnamon, almond, and olive oil using different combinations of DELs. This study's results indicate that DELs are far better in phenolic extraction efficiency than conventional solvents.[183] Garcia et al.[184] extracted different phenolics from olive oil using a DEL system containing ChCl as an HBA paired with various HBDs and the results showed that DEL based on ChCl and xylitol and ChCl paired with 1,2-propanediol showed superior results with remarkable extraction efficiency (30–70%) as compared to methanol and water (conventional solvents). Astonishing higher enrichment factor i.e., 444.59–516.58% of apigenin and luteolin, were observed[29] by applying ChCl:polyethylene glycol (1:8) diluted with 40% H_2O. These authors tested thirty four ChCl and betaine-based DEL combinations and found that betaine hydrochloride and propylene glycol form two different types of hydrogen bonds and water molecules play an important role in the interaction between HBD and HBA. Mansur et al.[185] tested the efficacy of eighteen combinations of DELs to extract flavonoids from buckwheat sprouts via UAE. It was observed that DEL containing ChCl and triethylene glycol (1:2) offered a higher extraction yield during UAE. The citrus peel has been ranked as a sustainable source of fiber and flavonoids. Using ultrasonic irradiation, various DEL combinations were prepared to extract flavonoids from citrus peel. The results again recommend that DEL containing ChCl, N-methyl urea, and levulinic acid can exhibit higher recovery rates than conventional solvents.[186]

Paradiso et al.[187] prepared a NaDEL based on glucose and lactic acid with the addition of water to reduce the viscosity for extraction of tyrosol derivatives from olive oil. These authors also declared lactic acid and glucose-based NaDELs as green and benign solvents[187] while t he available experimental data record is not sufficient to announce the above-mentioned DEL system as biodegradable. Moreover, the majority of researchers applying NaDELs as extraction solvent have not undertaken isolation of NaDELs from the final extracts produced. Liu et al.[121] established a novel method for the extraction and quantification of lignins from sesame oils using deep eutectic liquids (DELs) under UALLME and observed that DELs comprising ChCl and alcohols can efficiently liberate lignins (both polar and non-polar) as compared to conventional solvents. A similar extraction setup has been also used for the extraction of phenolic antioxidants and counter-validated for synthetic antioxidants. It was observed that DEL made up of ChCl and EG in a molar ratio of 1:2 exhibited the highest extraction efficiency.[188]

Likewise, NaDELs comprising ChCl and ascorbic acid also provided the best overall extraction results as compared to conventional organic solvents as well as other DEL combinations.[189] A hydrophilic DEL comprising ChCl and EG was also applied during vortex-assisted liquid-liquid extraction of alkyl gallates antioxidants from edible oils. It was reported that the DEL phase becomes rich in phenolic antioxidants in the case of hydrophilic DEL combinations.[190]

Microwave-assisted extraction (MAE)

Bubalo et al.[191] compared microwave-assisted extraction (MAE) with ultrasound-assisted extraction (UAE) while applying DEL as an extraction solvent. The results showed that a deep eutectic mixture ChCl and oxalic acid with a small amount of water exhibited superior extraction efficiency of phenolics from grape skin when coupled with microwave or ultrasound pretreatments. Panić et al.[192] prepared ready-to-use phenolic extracts containing NaDELs based on ChCl and citric acid from various food by-products via coupling of microwave and ultrasound irradiation.[192] Huang et al.[139] prepared thirteen different combinations of NaDELs using ChCl, Glycerol, sugars, and amino acids and applied them for the extraction of rutin in combination with water and ultrasound irradiation. The authors claimed that the majority of prepared NaDELs were nontoxic and biodegradable. Panić et al.[122] utilized microwave and ultrasound irradiation to extract flavonoids from grape pomace using different NaDELs. The results showed that NaDELs based on choline chloride and citric acid diluted with 30% water were most effective in terms of extraction efficiency.

Ferrone et al.[193] prepared some other combinations to extract various bioactive compounds from vegetable oil. Six targeted compounds were extracted and quantified from oils using a DEL based on dimethylglycine and phenylacetic acid.[193] Jeong et al.[194] introduced a single step-DEL preparation to extract bioactives from peppermint (a non-caffeinated herbal tea). The results of this study indicate that DEL consisting of ChCl and glucose in a molar ratio of 5:2

exhibited the best results in terms of extraction efficiency as compared to other DEL combinations as well as conventional solvents.[194] Park et al.[195] extracted various phenolic acids from *Herba artemisiae* using a binary mixture of methanol in water and tweleve different combinations of DEL. The results showed that DEL based on urea and tetramethylammonium chloride showed remarkably greater extraction efficiency than other solvent mixtures.[195] It was observed that all the tested DEL combinations were more efficient than the conventional solvents while a DEL made up of ChCl and levulinic acid was more tailorable and effective.[196]

Nemati et al.[197] applied the ChCl:hexanedioic acid as DEL and ion-pairing agent during the microwave-assisted extraction of tuna fish amines (histamine, tryptamine, tyramine, and dopamine). The authors claim the ChCl:hexanedioic acid facilitates the dispersion of amines during liquid-liquid micro-extraction. Even though the synthesis of DELs has been claimed to take place in the aqueous medium, it was speculated that one end to carboxylic acid participates in the formation of DEL while the other can form ion pair with analyte deep eutectic solvent while the other one is used to form ion pair with the analytes. The ion-pairing or extraction efficiency can be further regulated by irradiation, pH, and salt concentration.

Solid phase extraction (SPE)

Recently, Zhang et al.[198] applied ChCl-based DEL as a stationary phase for purifying lysine-acetylated peptides of MDCplam, GGAKacR, GGAKR, and VYALKglu sequences. The authors observed that incorporating ChCl and methacrylic acid (1:2) based DEL during synthesizing molecularly imprinted Monolith can improve the identified peptides and proteins. The elution of bound protein with a solvent comprising acetic acid, water, and acetonitrile increased the protein recovery up to 3.9 folds (Table 2). In another study,[199] silica gel was impregnated with DEL made up of tetraethylammonium chloride (TEAC) and ethylene glycol (EG) in 1:2 and utilized for the solid phase dispersion of triazine herbicides from brown sugar. Finally, TEAC and EG comprisin DEL offered recoveries up to 95.0–101.7% for six triazines.Guo et al.[200] prepared DEL-assisted methomyl (a carbamate insecticide) containing carbon quantum dots (CQDs) and tested them as Acetylcholinesterase (AchE) inhibitors. The scheme was subsequently applied for the determination of methomyl in various cereal grains.

El Kantar et al.[201] developed a novel method involving high voltage electrical discharge followed by solid-liquid extraction of phenolic compounds from grapefruit peels using NaDELs based on glucose and lactic acid. The results showed that extraction carried out with NaDELs and electrical discharge as pretreatment showed superior results compared to conventional techniques and solvents.[201] Li et al.[204] have observed that the nature of DEL, adsorbent/sample ratio, nature, and amount of eluent could affect the extraction efficiency. Wang et al.[202] introduced a novel method for detecting Rhodamine B in various food samples using different combinations of DEL utilizing UPLC for quantification. The results showed that DEL based on choline chloride and ethylene glycol in a molar ratio of 1:3 was most effective in terms of the extraction efficiency of Rhodamine B. Various pharmaceutical products and drugs exhibit harmful impacts when they become a part of the aquatic environment and accumulate in aquatic organisms. Liu et al.[203] introduced a method for extracting and chemical characterization of various personal care and pharmaceutical products and fish oils using DELs. Another carcinogenic mycotoxin, aflatoxins was determined by Wu et al.[204] using a solid-phase dispersion method. The authors applied three combinations of DEL to determine aflatoxins in different crops and found that DEL consisting of tertiary-butyl acetate and hexyl alcohol was most effective in recovering the targeted compounds.

The advent of solid-phase microextraction (SPME), has produced a torrent of pioneering work regarding the use of MDELs containing MOF, graphene, nanomaterials, and molecularly imprinted polymers as sorbent for the extraction of metals, organic pollutants, and toxins. Karimi et al.,[205] first time applied MDELs made up of nanoparticles of Fe_3O_4 and ChCl:urea (1:1.25) for the legendless pre-concentration of lead and cadmium from environmental samples. The authors claimed that an aqueous solution of DEL system like ChCl-urea (1:2.5) combines with Fe_3O_4 nanoparticles it loses their water and become sorbed on the surface of nanoparticles via hydrogen bond or other electrostatic interactions and can bind with a variety of analytes.

Challenges and future prospects

Deep eutectic liquids (DELs) have become fascinating liquids of the 21st century for their biodegradability, low toxicity and cost, preparation easiness, and adoptable physicochemical properties. The most promising feature of these liquids rises for their tailorable nature i.e., a change in HBD/HBA ratios or moisture level can drastically change the melting point, viscosity, surface tension, and electrical conductance. For a similar reason, a great deal of research has become into practice to test the suitability of these liquids in various fields, including modification and synthesis of sorbents, electrochemistry, chromatography, electrophoresis, and various separation process with a particular focus on the extraction of bioactive compounds. In addition, their biodegradable nature with no toxicity and low reactivity with water makes them an ideal solvent for various analytical applications. Besides, their low-cost production with a great variety of readily available raw materials is the cherry on top.[5,135] Applying deep-eutectic liquids as an extraction solvent can lead the separation technology into green and sustainable analytical chemistry. However, it may take a couple of exhausting decades as these liquids face many logical and technical challenges.

First of all, the formation of DELs needs to be authenticated and carefully characterized as it does not involve any chemical reaction rather it happens due to reversible intermolecular interactions or entanglement between HDB and

HBA species. Another technical challenge is concerned with the thermal stability of these liquids. The available knowledge regarding the mechanism, formation, and characterization of DELs shows that the thermal stability of these liquids, particularly those prepared from natural HBDs and HBAs stands questionable.

Overall, mixing a deep eutectic liquid (DEL) with water often increases mobility, decreases viscosity and density, and improves the electrical conductance of the resultant solutions but the final mixtures may behave as binary or ternary solution rather than a deep-eutectic liquid. This may lead to completely astonishing/ambiguous outcomes and there is a prompt need to re-define deep-eutectic liquids. In the case of hydrophobic deep eutectic liquids, the water contamination or moisture tremendously affects the geometry and physicochemical features of the liquid formed. Some studies say that the addition of water up to 5% (w/w) can get absorbed in the molecular matrix of liquid and does not affect the physicochemical attributes of liquids. However, the further addition of water will dampen the intermolecular and intra-molecular interactions drastically. This variation can be further exploited to render these liquids more tailorable.

Another key hurdle regarding the use of DELs as extraction solvent raises for the relatively higher viscosities, densities, and surface tension of these liquids. In this context, many opportunities are available to design these DEL features of DELs by changing the stoichiometric ratios of HBDs and HBAs, mixing techniques, and molecular dynamics. These opportunities are at least more fascinating and more accessible than those available regarding ionic liquids. The simple but less sustainable way to optimize the physicochemical features of DELs for their application as extraction solvents involves mixing these liquids with conventional organic solvents or water. For example, Đorđević et al.[160] have observed that mixing of ChCl and Urea (1:2) or Triethanolamine and glycerol (1:2) based solvent with water or ethanol offered a higher extraction yield of phenolic compounds from black mustard seed. In another study, Wojeicchowski et al.[161] report that mixing ChCl and 1,2-propanediol-based DEL system with water (50:50 v/v) lowers its density and viscosity and improves the solubility of rosemary phenolics. Meanwhile, the molecular modeling data (COSMO-RS) prediction indicates that the addition of water to the above DEL system does not improve the solubility of analytes, which implies that dilution of that particular DEL with water has improved the mass-transfer rates not the solubilities. These trends need more comprehensive trials before reaching final conclusion.

The recycling/recovery of DEL system from the analytes/extracts exist as an unaddressed challenge. Many researchers claim that DELs especially NaDELs are recyclable but these claims lack sound technical and practical background. The dispersive type (DLLME) and salting-out assisted microextraction need to be further investigated for the intelligent recycling of DELs from the matrix. A magnetic recycling method has been reported to recycle ChCl and $FeCl_4$ based DEL systems from heptanal and hexanal extracted from edible oils.[150]

Conclusions

The search for novel green solvents has been the talk of the town since the mid of 20th century. Deep-eutectic liquids (DELs) have emerged as potential solvents for their characteristic and tunable features. Besides, the involvement of inexpensive and readily available HBDs and HBAs and easily scaleable development protocols, these liquids stand among the most studied materials in 21st century. The natural deep-eutectic liquids (NaDELs) for their biocompatibility and low toxicity are more potential candidates to serve as a sustainable alternative to conventional toxic solvents. In addition, the nonvolatile character of DEL allows them to be paired with analytical instruments like UV-Visible spectrophotometer and HPLC for further analysis. However, this very nonvolatile property of DEL offers an isolation challenge at the end of extraction or synthesis.

To put the whole discussion in a nutshell, it can be said that the future of DEL is quite auspicious as it has the potential to serve as a sustainable alternative for novel applications. However, to utilize DEL to its fullest potential, more coherent and systematic research need to be dedicated to intermolecular entanglement, radial distribution function, target-oriented molecular modeling, solvatochromic potential, recyclability, and designable physicochemical attributes of DEL systems. The recycling of DELs from extraction or reaction media needs more exhaustive and comprehensive studies to derive through a straight-way methodology and right now it would be fair to say that recycling or isolation of deep eutectic liquids after their application is still an unaddressed challenge regarding the utilization of these green liquids as solvents.

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